STA 445 S26 Assignment 5

2026-03-18

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.
b. Sept 13, 2010.
c. Sep 13, 2010.
d. S 13, 2010. Comment on the month abbreviation needs.
e. 07-Dec-1941.
f. 1-5-1998. Comment on why you might be wrong.
g. 21-5-1998. Comment on why you know you are correct.
h. 2020-May-5 10:30 am
i. 2020-May-5 10:30 am PDT (ex Seattle)
j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

```r
mdy("Sept 13, 2010", abbr = TRUE ) # year month day format does not use "sept" it uses SEP
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA NA
```

```r
mdy("S 13, 2010", abbr = TRUE ) # Same issue as above, it wants "sept"
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA NA
```

```r
dmy("07-Dec-1941")
```

```
## [1] "1941-12-07"
```

```r
dmy("1-5-1998") # it's not clear which of these is intended to be the month and ay
```

```
## [1] "1998-05-01"
```

```r
dmy("21-5-1998") #there is no 21st month so there is no ambiguity as to which of these is the day
```

```
## [1] "1998-05-21"
```

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

```r
ymd_hm("2020-May-5 10:30 am", tz = "PST8PDT")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

```r
ymd_hm("2020-May-5 10:30 am", tz = "America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

```r
birthday <- mdy("09-16-2003")
today <- mdy("03-18-2026")
```

a. Calculate the date of your 64th birthday.

```r
birthday + years(64)
```

```
## [1] "2067-09-16"
```

b. Calculate your current age (in years).

```r
yearsalive <- as.period(birthday %--% today, unit = "year") %>% year()
yearsalive
```

```
## [1] 22
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
nextbirthday <- birthday + years(yearsalive)
nextbirthday
```

```
## [1] "2025-09-16"
```

d. The number of *days* until your next birthday.

```r
as.period(nextbirthday %--% today, "days")
```

```
## [1] "183d 0H 0M 0S"
```

e. The number of *months* and *days* until your next birthday.

```r
as.period(nextbirthday %--% today, "months")
```

```
## [1] "6m 2d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
call_time <- mdy_hm("05-08-2015 10:30am", tz = "America/Phoenix")
with_tz(call_time, "Pacific/Auckland")
```

```
## [1] "2015-05-09 05:30:00 NZST"
```

Problem 4

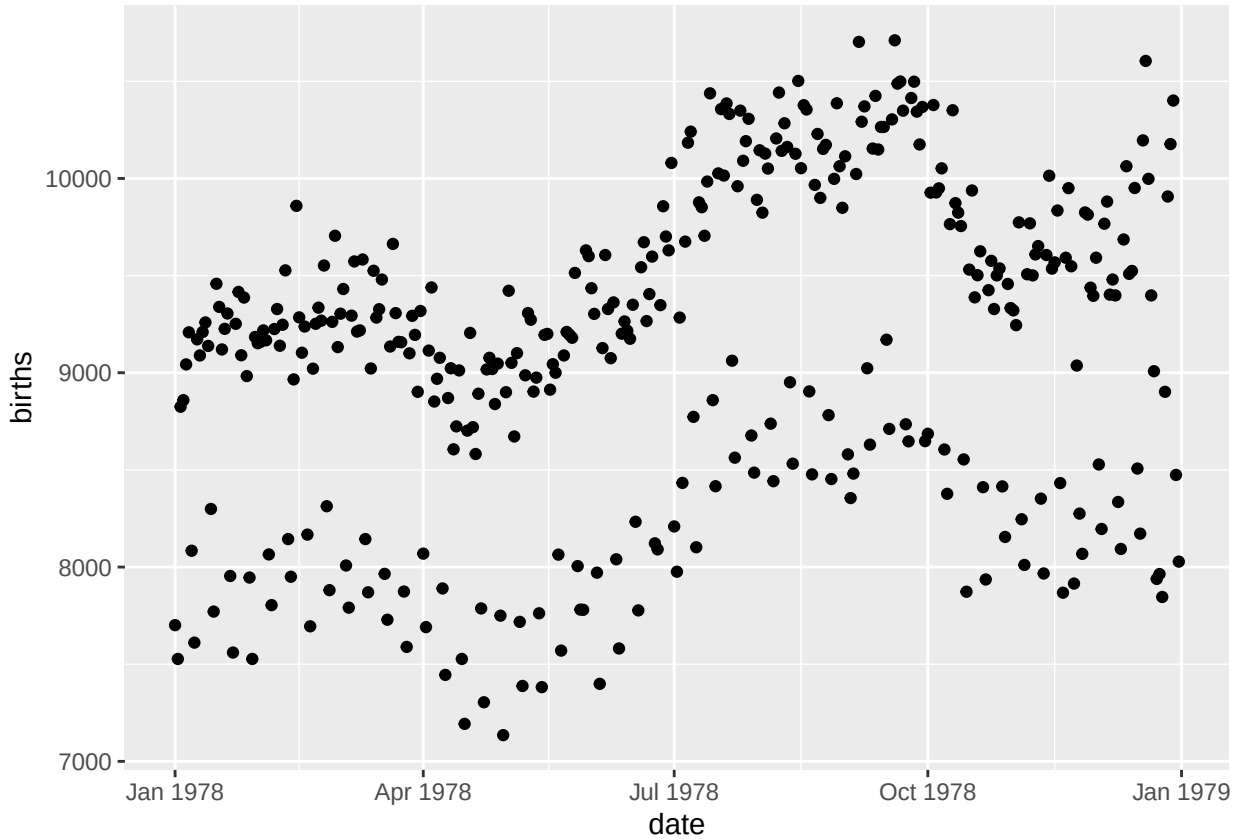
It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```r
birthdata <- data.frame(Births78) %>% select("births", "date")
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```r
ggplot(birthdata, aes(x = date, y = births)) +
  geom_point()
```



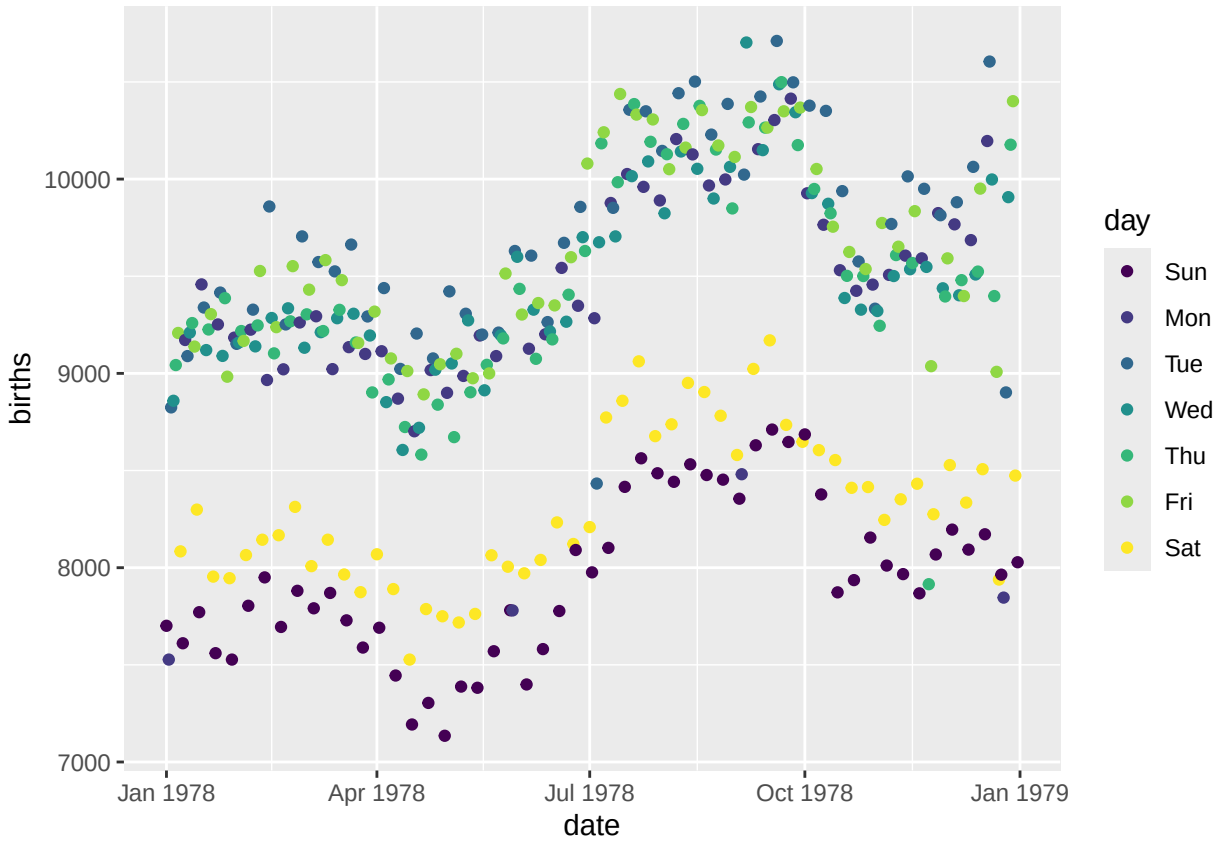
there is a trend here where we effectivly have two different, nearly ideticle lines which indicates that some very regular event is dropping the birth rate. It's hard to tell at this scale but I suspect that fewer babies are born on weekends since that could come into play when medical intervention is involved

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```r
birthdata_dow <- mutate(birthdata, day = wday(date,label= TRUE, abbr=TRUE))
```

d. Plot the data with the point color being determined by the day of the week variable.

```r
ggplot(birthdata_dow, aes(x = date, y = births, color = day)) +
  geom_point()
```



indeed saturday and sunday are the lower days